removing words, or rephrasing sentences—are enough to alter the decoding trajectory of the LLM and make the original seed unrecoverable. These edits are commonplace in communication and do not affect semantic meaning, yet they completely destroy the hidden information. Thus, like the earlier rejection sampling schemes, Meteor fails to support robust steganography.

Similar vulnerabilities appear in other systems that embed covert data into low-level features of ciphertexts or generated text. Wink [10] is a system that embeds hidden messages within the random values of encrypted communications, achieving deniability by ensuring that ciphertexts can plausibly be interpreted in multiple ways. However, like Horel et al.'s construction, Wink is fragile in the face of re-randomization attacks: an adversary can simply decrypt and re-encrypt a message, thereby removing any embedded content without altering the plaintext. Since these re-randomized ciphertexts remain valid and indistinguishable from ordinary ones, this attack is undetectable and does not interfere with honest communication.

Other approaches in linguistic steganography, such as SegFree [40], attempt to embed covert messages directly within natural language output by using generative methods. Additionally, recent work such as Discop [14] has explored provably secure steganographic methods that ensure covertext distributions remain indistinguishable from specified original distributions, highlighting the importance of statistical matching in steganography.

The broader literature on watermarking LLM outputs provides further evidence of how fragile these systems can be. Recently established impossibility results for strong watermarking in generative models show that certain watermarking schemes can be removed without significant degradation to the generated content [44]. However, these results do not apply to our system, as our watermarking-based approach has stricter requirements than the general case.

Christ and Gunn [13], done concurrently to this work, use watermarks to embed hidden bits in text via pseudorandom error-correcting codes, a technique similar to our multi-watermark embedding method. Zamir [43], done concurrently to this work, further extends this and presents a steganographic method embedding hidden messages in LLM outputs while ensuring undetectability, which is similar to our multi-watermark embedding approach. Crucially, it is not robust to paraphrasing text. Our embeddings based approach is robust to paraphrasing attacks.

Taken together, the prior work discussed here paints a clear picture: the state of the art in steganography—across ciphertexts, language, and watermarking—remains vulnerable to simple, realistic tampering. Existing systems emphasize undetectability and decoding under ideal conditions but offer little protection against covertext modifications. Robustness—the ability to recover hidden messages after adversarial edits that preserve semantics—has received scant attention and is largely absent from formal definitions and constructions. In this work, we initiate a systematic study of robustness in steganographic systems and demonstrate that meaningful progress is possible in both theory and practice.

3 Preliminaries

3.1 Pseudorandom Functions

We use the following standard notion of a pseudorandom function. Let $\mathsf{PRF} = (\mathsf{PRF.Gen}, \mathsf{PRF.Eval})$ be a function family over domain $\{\mathcal{X}_\lambda\}_{\lambda\in\mathbb{N}}$ with range $\{\mathcal{Y}_\lambda\}_{\lambda\in\mathbb{N}}$ and key space $\{\mathcal{K}_\lambda\}_{\lambda\in\mathbb{N}}$, such that:

- $\mathsf{PRF.Gen}$ is a probabilistic polynomial-time algorithm, which takes as input the security parameter $\lambda \in \mathbb{N}$ and outputs a key $k \in \mathcal{K}_\lambda$.

– $\mathsf{PRF.Eval}$ is a deterministic polynomial-time algorithm, which takes as input a key $k \in \mathcal{K}_\lambda$ and a domain element $x \in \mathcal{X}_\lambda$ and outputs a value $y \in \mathcal{Y}_\lambda$.

For ease of notation, for a key $k \in \mathcal{K}_\lambda$, we denote by $\mathsf{PRF}_K(\cdot)$ the function $\mathsf{PRF.Eval}(K, \cdot)$. We also assume without loss of generality that for every $\lambda \in \mathbb{N}$, it holds that $\mathcal{K}_\lambda = \{0,1\}^\lambda$ and that $\mathsf{PRF.Gen}(1^\lambda)$ simply samples k from $\{0,1\}^\lambda$ uniformly at random. Using these conventions, the following definition captures the standard notion of a pseudorandom function family.

Definition 1. *A function family* $\mathsf{PRF} = (\mathsf{PRF.Gen}, \mathsf{PRF.Eval})$ *is* pseudorandom *if for every probabilistic polynomial-time algorithm D, there exists a negligible function $\nu(\cdot)$ such that*

$$\mathsf{Adv}^{\mathrm{prf}}_{D,\mathsf{PRF}}(\lambda) := \left| \Pr_{k \leftarrow \{0,1\}^\lambda} \left[\mathsf{D}(1^\lambda)^{\mathsf{PRF}_k(\cdot)} = 1 \right] - \Pr_{f \leftarrow \mathcal{F}_\lambda} \left[\mathsf{D}(1^\lambda)^{f(\cdot)} = 1 \right] \right| \leq \nu(\lambda),$$

for all sufficiently large $\lambda \in \mathbb{N}$, where $\mathcal{F}_\lambda$ is the set of all functions mapping $\mathcal{X}_\lambda$ into $\mathcal{Y}_\lambda$.

3.2 Language Models

A **language model** is a pair $(\mathcal{T}, M)$, where $\mathcal{T} = \{t_1, \ldots, t_n\}$ is a finite set of symbols called *tokens*, and M is a probabilistic polynomial-time algorithm. The set $\mathcal{T}$ includes the special termination symbol ter. M takes as input a *prompt* $\mathsf{prompt} \in \mathcal{T}^*$ and a (possibly empty) sequence $\boldsymbol{w} = (w_1, \ldots, w_{i-1}) \in \mathcal{T}^{i-1}$ of tokens.[1] On this input, M outputs a probability distribution $P_i \leftarrow M(\mathsf{prompt}, \boldsymbol{w})$ over $\mathcal{T}$. That is, for every $t_j \in \mathcal{T}$, $P_i(t_j) \in [0,1]$ and $\sum_{j \in [n]} P_i(t_j) = 1$.

For a language model $(\mathcal{T}, M)$ and a prompt $\mathsf{prompt} \in \mathcal{T}^*$, we may consider the following probability distribution over $\mathcal{T}^*$:

1. Compute $P_1 \leftarrow M(\mathsf{prompt})$, and sample $w_1 \leftarrow\$ P_1$. If $w_1 = \mathsf{ter}$, then output w_1 and terminate.
2. Set $i \leftarrow 1$ and $\boldsymbol{w} \leftarrow (w_1)$.
3. Until $w_i = \mathsf{ter}$ do:
   (a) Increment $i \leftarrow i + 1$.
   (b) Compute $P_i \leftarrow M(\mathsf{prompt}, \boldsymbol{w})$, and sample $w_i \leftarrow\$ P_i$.
   (c) Append w_i to $\boldsymbol{w}$.
4. Output $\boldsymbol{w}$.

We denote the process of sampling from this distribution by $\boldsymbol{w} \leftarrow\$ M^*(\mathsf{prompt})$, and call $\boldsymbol{w}$ a *response* of M on prompt.

3.3 Locally Sensitive Hashing

Locally Sensitive Hashing (LSH) is a technique used to map high-dimensional data into a lower-dimensional space while preserving proximity relationships between data points. Formally, a family of hash functions $\mathcal{H} = \{h : \mathbb{R}^d \to \{0,1\}^n\}$ is called (r_1, r_2, p_1, p_2)-sensitive for a similarity function $S : \mathbb{R}^d \times \mathbb{R}^d \to [0,1]$ if, for any $x, y \in \mathbb{R}^d$:

– If $S(x,y) \geq r_1$, then $\Pr[h(x) = h(y)] \geq p_1$,
– If $S(x,y) \leq r_2$, then $\Pr[h(x) = h(y)] \leq p_2$,

[1] For ease of presentation, we assume that $\mathsf{prompt} \in \mathcal{T}^*$. However, everything in this paper readily extends to the case where the prompt may contain symbols not in $\mathcal{T}$.

where $r_1 > r_2$ and $p_1 > p_2$.

The output of an LSH function is an n-bit binary string, where each bit is determined based on the properties of the hash function and the similarity of the input data points. LSH is commonly used in applications requiring approximate nearest neighbor search, as it efficiently maps similar inputs to similar hash outputs with high probability.

3.4 Embeddings

Embeddings are dense vector representations of objects, such as words, sentences, or paragraphs, in a high-dimensional space. An embedding function $E : \mathcal{T}^* \to \mathbb{R}^d$ maps a sequence of tokens $t_1, t_2, \ldots, t_n$ to a vector in $\mathbb{R}^d$, where d is the dimensionality of the embedding space.

Embeddings encode semantic and contextual information about the objects they represent. For example, embeddings generated by language models often reflect both lexical and syntactic relationships, enabling comparisons between similar or related objects. Formally, given two sequences $x, y \in \mathcal{T}^*$, their similarity can often be evaluated using geometric measures such as cosine similarity or Euclidean distance between their embeddings.

The embedding space is characterized by its high dimensionality and structure, which enables it to represent a wide range of linguistic and semantic features compactly and effectively.

4 Threat Model

Robust steganographic schemes must withstand adversaries capable of intercepting and modifying communications. In this section, we define the adversary's goals, capabilities, and constraints. To reflect different levels of adversarial sophistication, we introduce two threat models—a **weak model** and a **strong model**—that motivate our subsequent definitions of robustness.

4.1 Adversary Motivation and Justification

The adversary seeks to maintain control over information flow by targeting hidden communication. Their primary objectives are:

- **Detection**: Identifying whether a message contains hidden information, enabling punitive or censorial actions.
- **Disruption**: Modifying the message in a way that prevents successful decoding of hidden information by the intended recipient while leaving the meaning of the message intact.

However, the adversary must balance their actions against the potential for societal backlash. Excessive tampering that fundamentally disrupts the intent or meaning of messages can lead to public discontent or retaliation. Complete censorship of communication has been found to increase protests [32] and be prohibitively expensive [30]. Governments have been found to deploy selective censorship to achieve particular goals such as stability instead of censoring anything critical of the government [23]. For example:

- If citizens notice that their everyday communications are being significantly altered or misrepresented, trust in the communication system may collapse.
- Such disruptions could incite rebellion or other forms of resistance, undermining the adversary's goals.